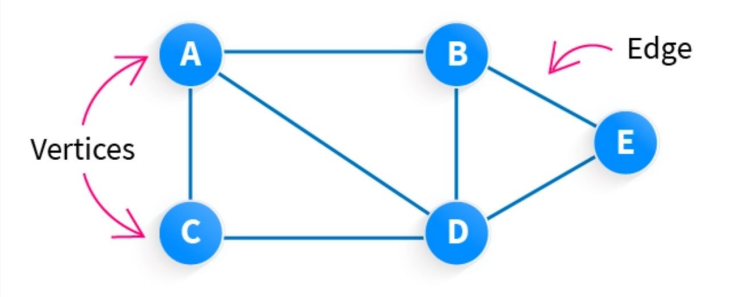
Graph and its representations

Graph

A Graph is a non-linear data structure consisting of vertices and edges.

The vertices are sometimes also referred to as nodes and the edges are lines or arcs that connect any two nodes in the graph.

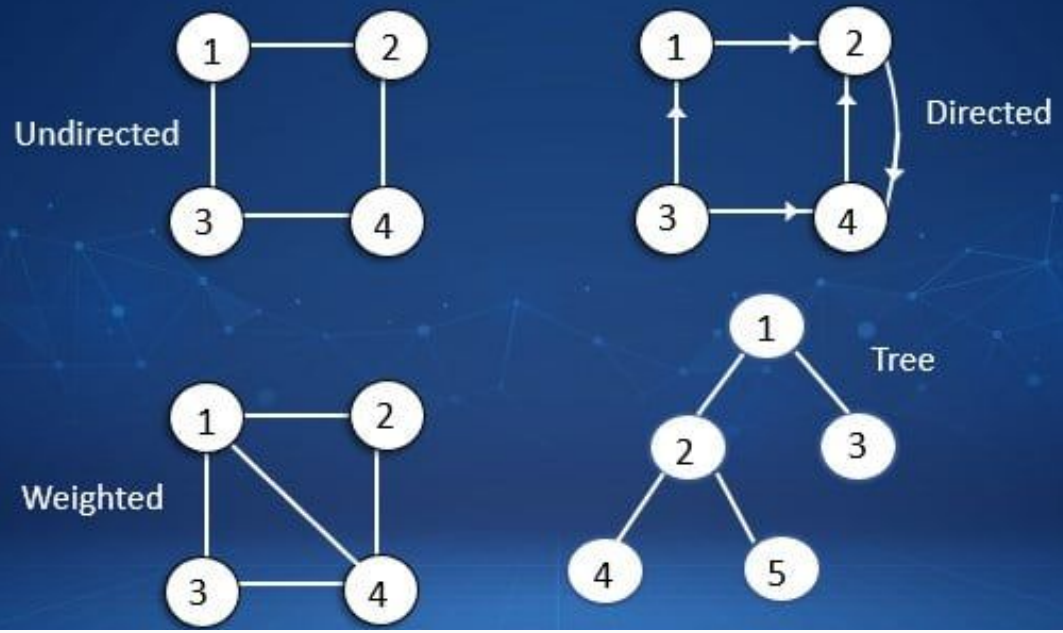


The graph is denoted by G(V, E).

V = { A, B, C, D, E}
E = { (A,B), (A,C), (A,D), (B,E), (B,D), (C,D), (D,E) }

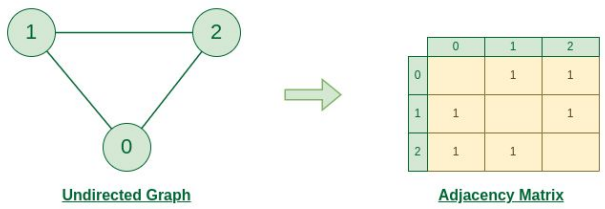
Note: There is a multiple neighbour of single point

Graph Representation



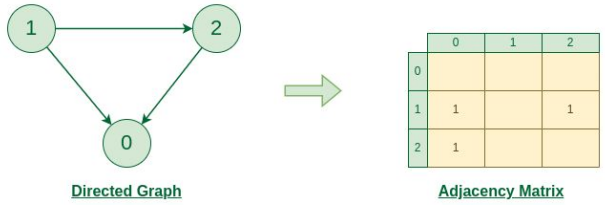
Representations of Graph

Adjacency Matrix



	0	1	2
0		1	1
1	1		1
2	1	1	

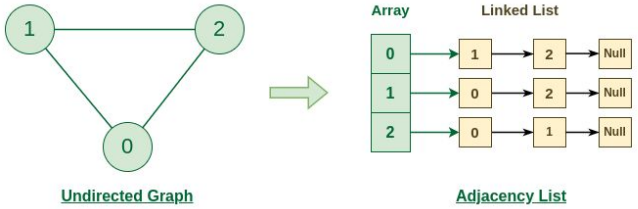
Graph Representation of Undirected graph to Adjacency Matrix



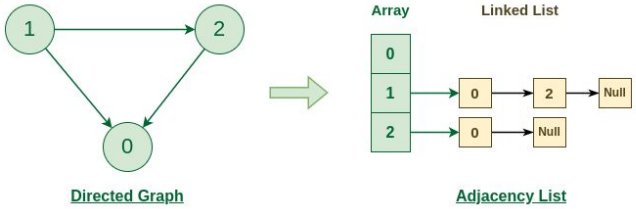
	0	1	2
0			
1	1		1
2	1		

Graph Representation of Directed graph to Adjacency Matrix

Adjacency List

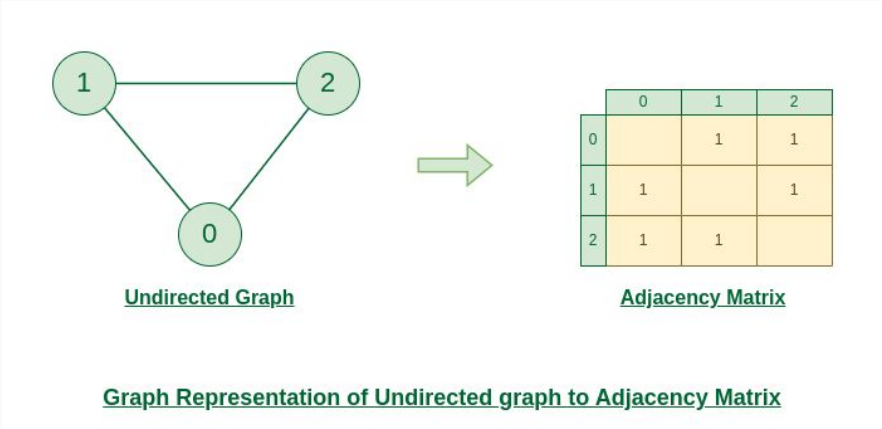


Graph Representation of Undirected graph to Adjacency List



Graph Representation of Directed graph to Adjacency List

Adjacency Matrix



Graph Representation of Undirected graph to Adjacency Matrix

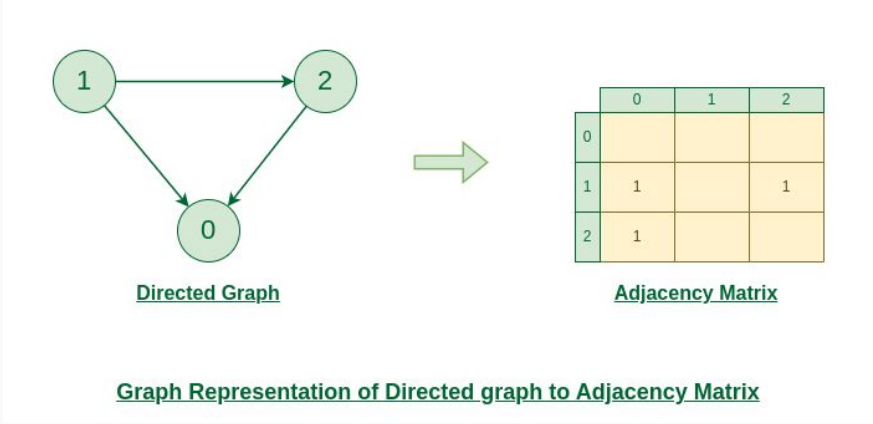
The figure shows an undirected graph.

Initially, the entire Matrix is initialized to 0.

If there is an edge from source to destination, we insert 1 to both cases (**adjMat[source][destination]** and **adjMat[destination][source]**) because we can go either way.

Coding…..

Adjacency Matrix



Graph Representation of Directed graph to Adjacency Matrix

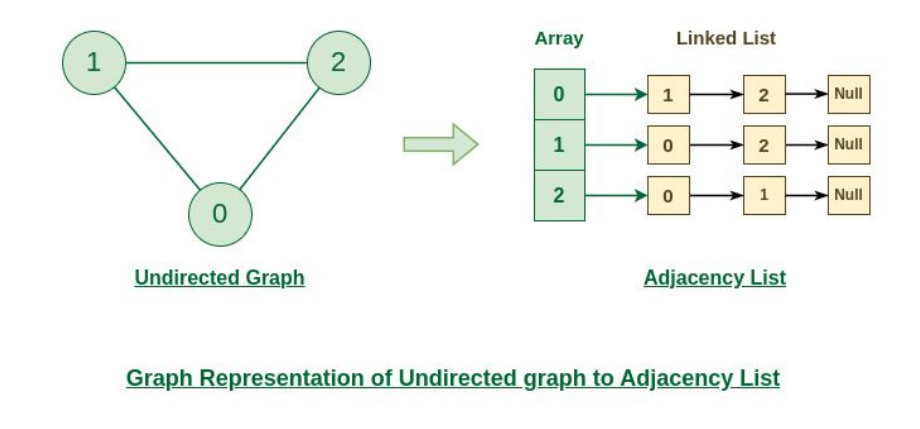
The below figure shows a directed graph.

Initially, the entire Matrix is initialized to 0.

If there is an edge from source to destination, we insert 1 for that particular **adjMat[source][destination]**.

Coding…..

Adjacency List



Graph Representation of Undirected graph to Adjacency List

The below undirected graph has 3 vertices.

So, an array of list will be created of size 3, where each indices represent the vertices.

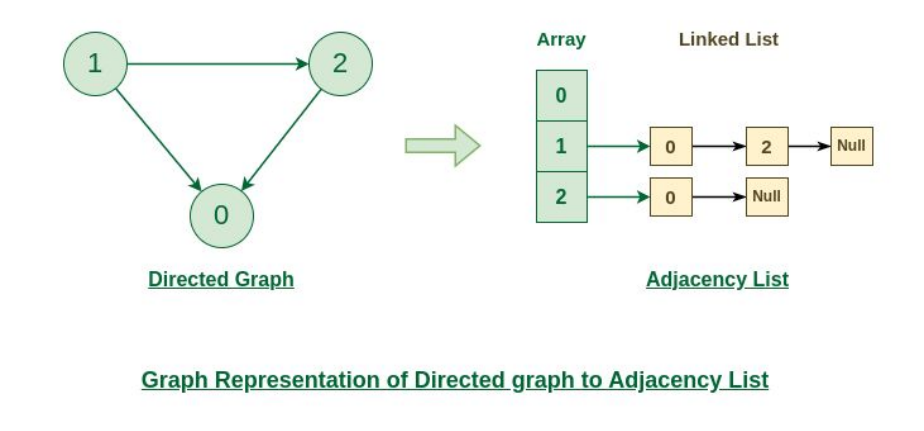
Now, vertex 0 has two neighbours (i.e, 1 and 2). So, insert vertex 1 and 2 at indices 0 of array.

Similarly, For vertex 1, it has two neighbour (i.e, 2 and 0) So, insert vertices 2 and 0 at indices 1 of array.

Similarly, for vertex 2, insert its neighbours in array of list.

Coding…..

Adjacency List



The below directed graph has 3 vertices.

So, an array of list will be created of size 3, where each indices represent the vertices.

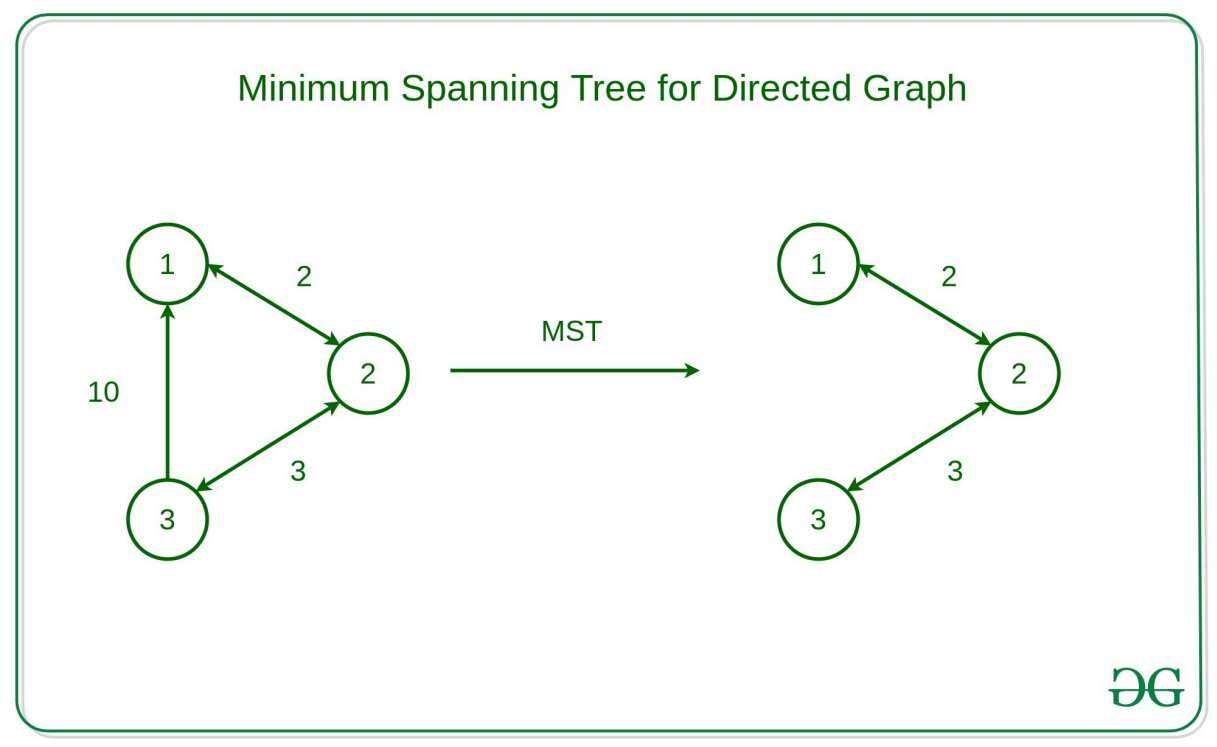
Now, vertex 0 has no neighbours.

For vertex 1, it has two neighbour (i.e, 0 and 2). So, insert vertices 0 and 2 at indices 1 of array.

Similarly, for vertex 2, insert its neighbours in array of list.

Coding…..

What is Minimum Spanning Tree (MST)



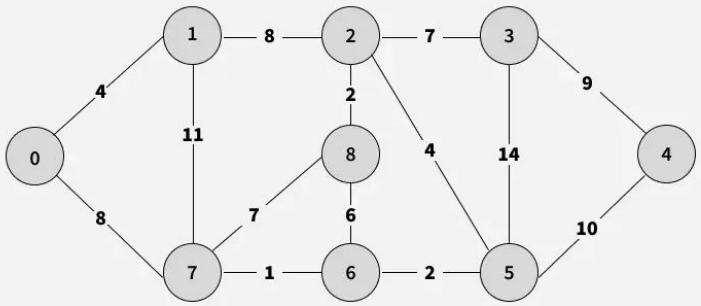
PRIMS Algorithms

Prim's algorithm is a **greedy algorithm** used to find the **Minimum Spanning Tree (MST)** of a connected, weighted, undirected graph.

- **Input:** A weighted, connected, undirected graph.
- **Output:** A tree that connects all vertices with the minimum possible total edge weight.

The idea is:

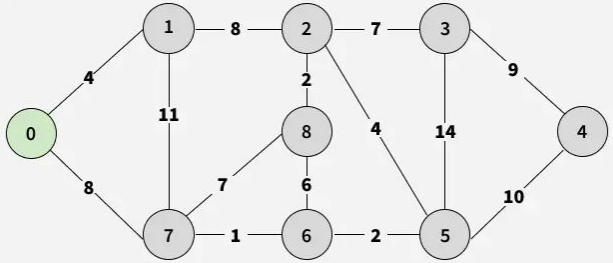
1. Start with any vertex.
2. Repeatedly add the **smallest edge** that connects a visited vertex to an unvisited vertex.
3. Continue until all vertices are included.



Prim's Algorithm

01 Step

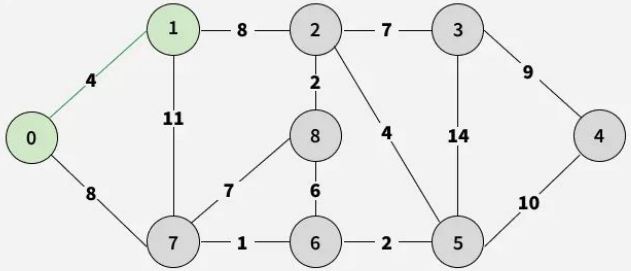
Start with edges {0,1} and {0,7}; pick the minimum {0,1} and add vertex 1.



Prim's Algorithm

02 Step

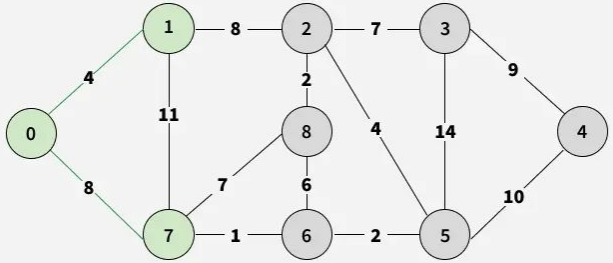
Choose between {0,7} and {1,2}; pick {0,7} (or {1,2}), adding vertex 7.



Prim's Algorithm

03 Step

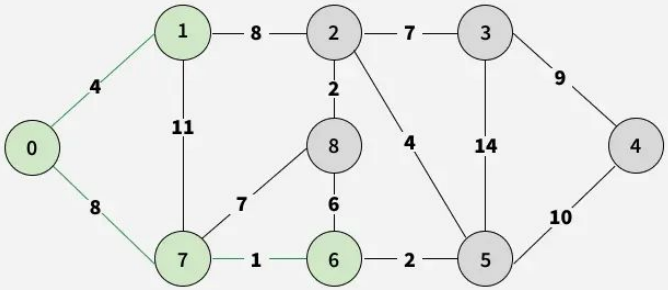
Select {7,6} as it has the least weight (1), adding vertex 6.



Prim's Algorithm

04 Step

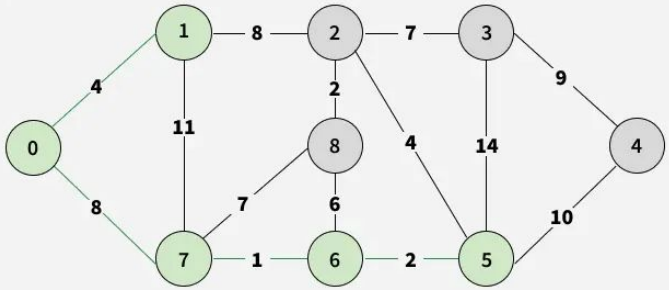
Among {7,8}, {6,8}, and {6,5}, pick {6,5} (weight 2) and add vertex 5.



Prim's Algorithm

05 Step

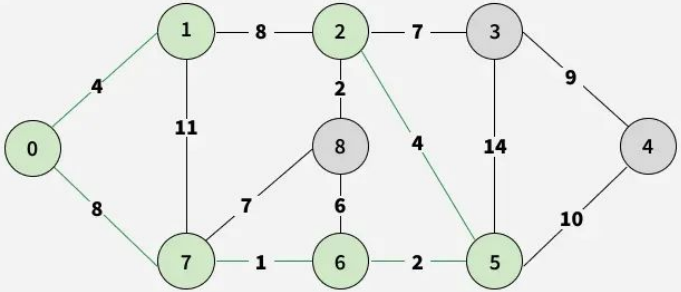
Select {5,2} (weight 4), adding vertex 2.



Prim's Algorithm

06 Step

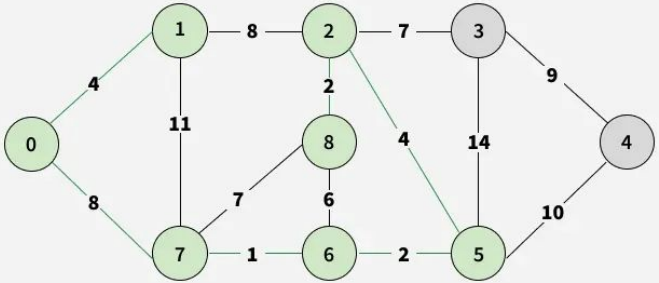
Choose {2,8} (weight 2), adding vertex 8.



Prim's Algorithm

07 Step

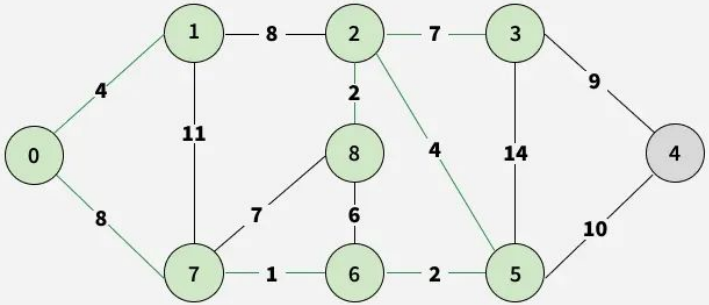
Pick {2,3} as the smallest edge, adding vertex 3.



Prim's Algorithm

08 Step

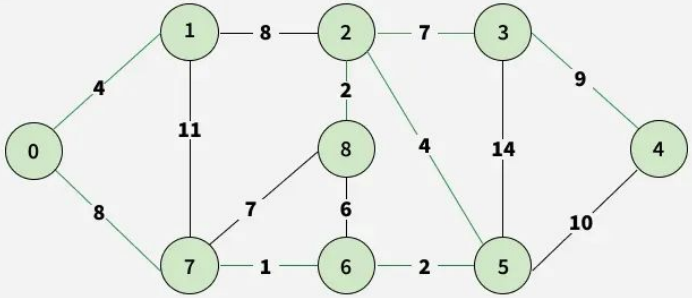
Finally, select {3,4}, adding vertex 4.



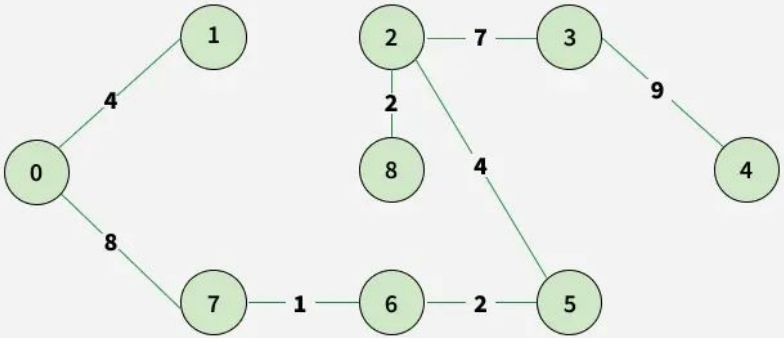
Prim's Algorithm

09 Step

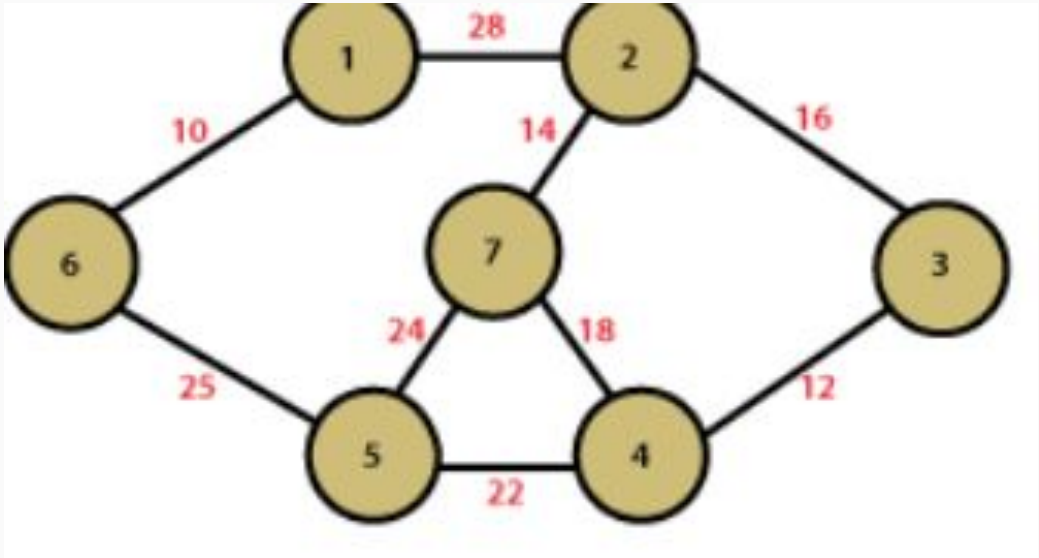
All vertices are included, so the MST is complete.



Prim's Algorithm



Prim's Algorithm


1
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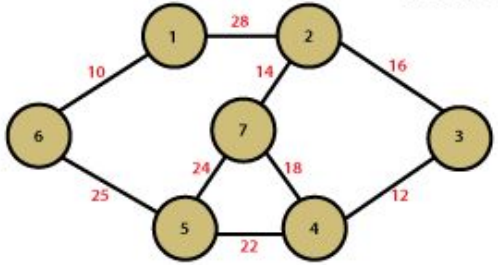
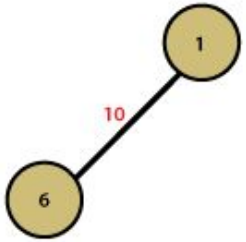
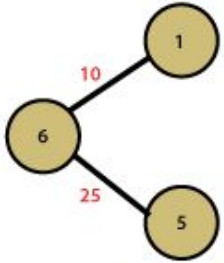
12
22

Prim's Algorithm

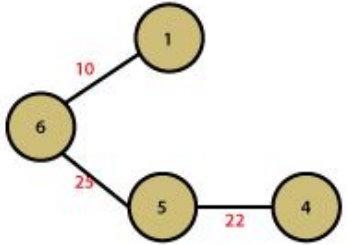




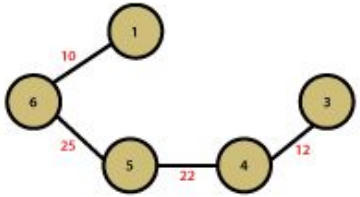
Step 1



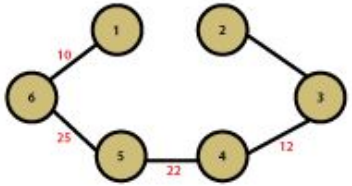
Step 2



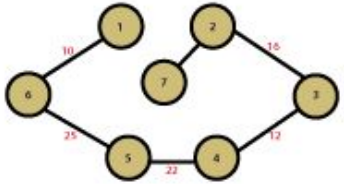
Step 3



Step 4



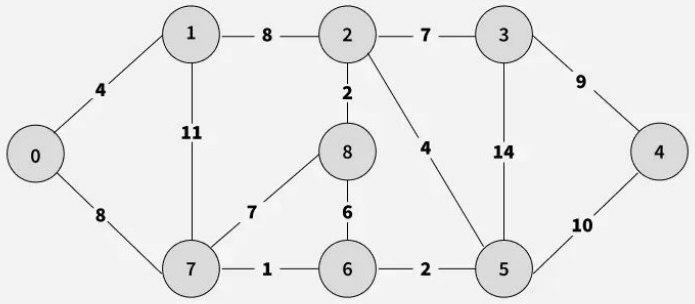
Step 5



Step 6

How to find MST using Kruskal's algorithm?

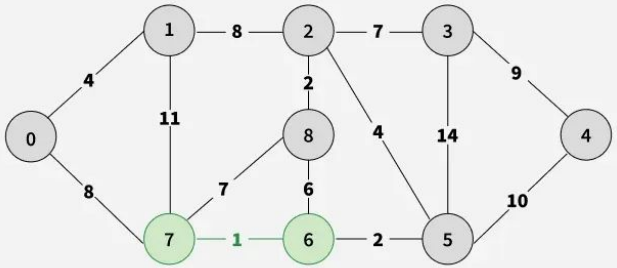
- Sort all the edges in a non-decreasing order of their weight.
- Pick the smallest edge. Check if it forms a cycle with the spanning tree formed so far. If the cycle is not formed, include this edge. Else, discard it.
- Repeat step 2 until there are (V-1) edges in the spanning tree.



Input Graph

02 Step

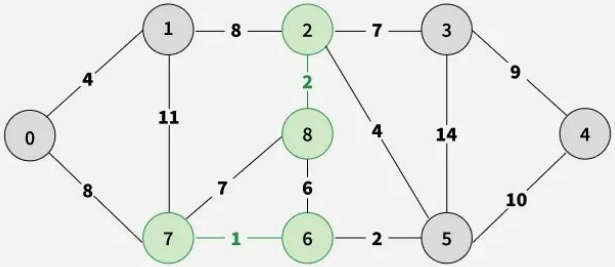
Pick all sorted edges one by one.
Pick edge 7-6. No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

03 Step

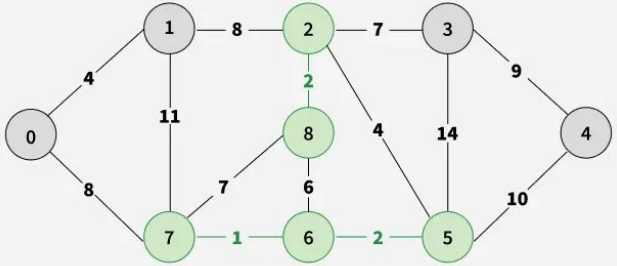
Pick edge 8-2. No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

04 Step

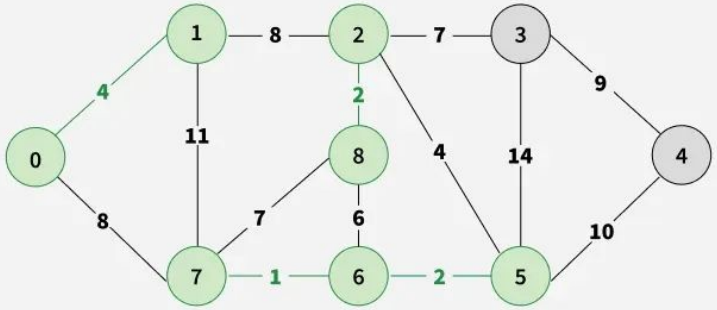
Pick edge 6-5. No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

05 Step

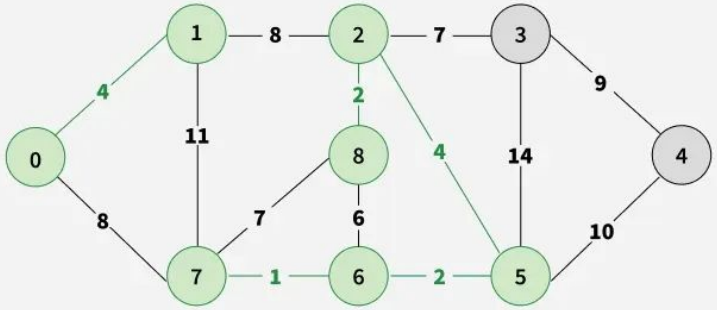
Pick edge 0-1. No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

06 Step

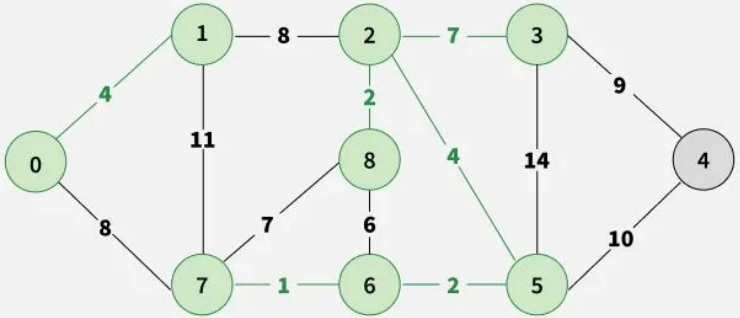
Pick edge 2-5. No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

07 Step

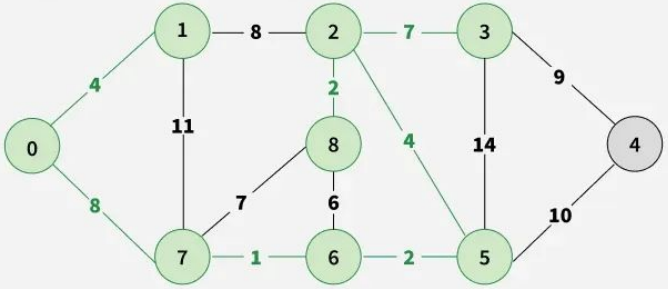
Pick edge 8-6. Since including this edge results in the cycle, discard it.
Pick edge 2-3: No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

08 Step

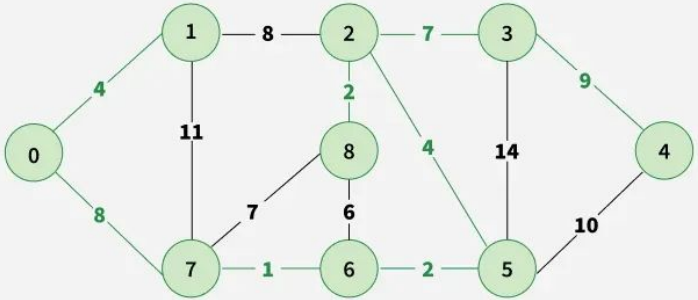
Pick edge 7-8. Since including this edge results in the cycle, discard it.
Pick edge 0-7. No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

09 Step

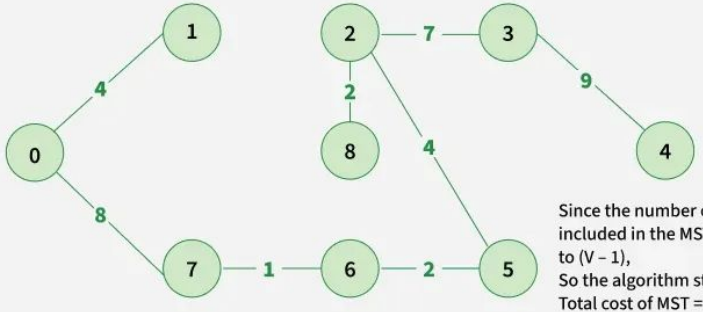
Pick edge 1-2. Since including this edge results in the cycle, discard it.
Pick edge 3-4. No cycle is formed, include it.



Kruskal's Minimum Spanning Tree (MST) Algorithm

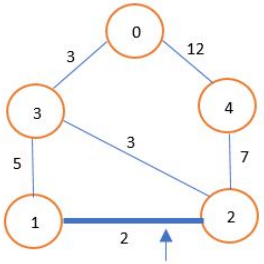
10 Step

Final obtained graph minimum spanning tree.



Since the number of edges included in the MST equals to (V – 1),
So the algorithm stops here.
Total cost of MST = (37)

Kruskal's Minimum Spanning Tree (MST) Algorithm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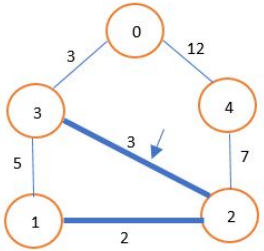
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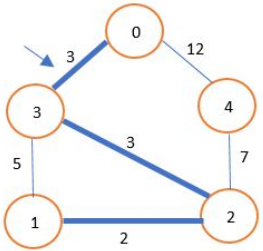
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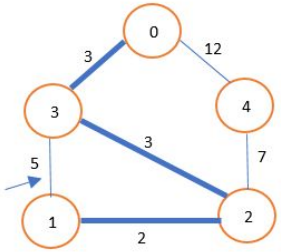
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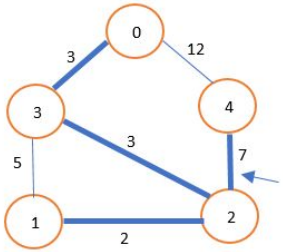
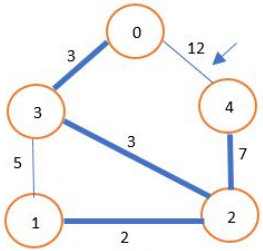
2

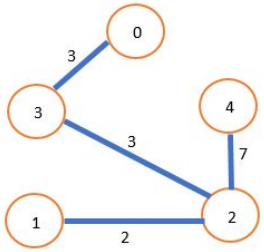
4


0
3
12
3
4
3
5
7
1
2
2

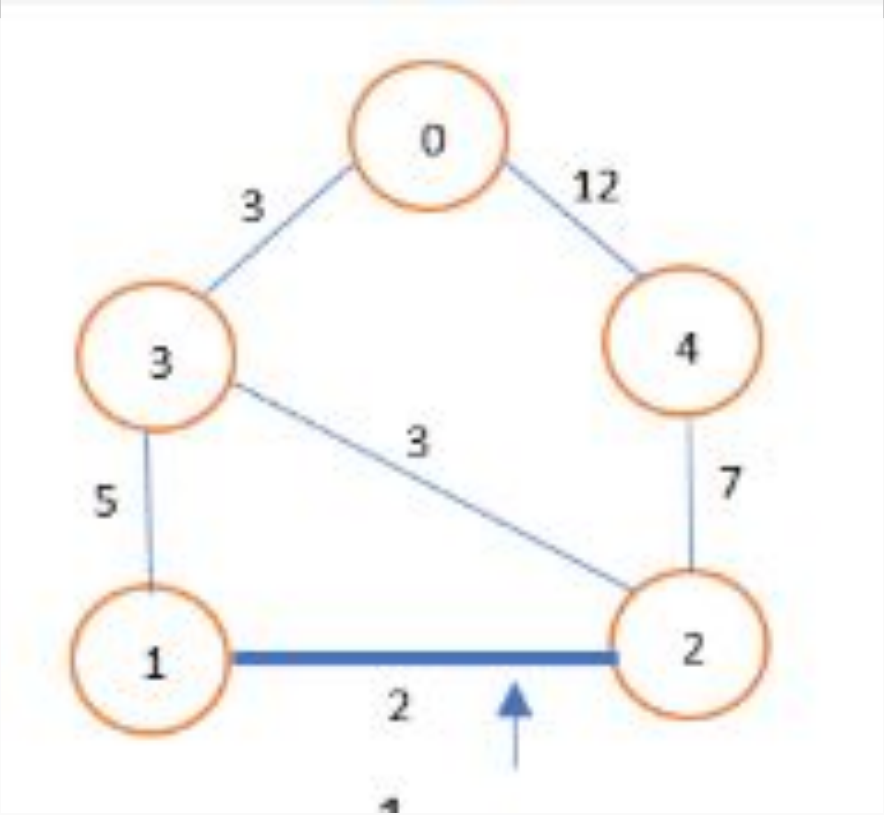
5


0
3
12
3
4
3
5
7
1
2
2

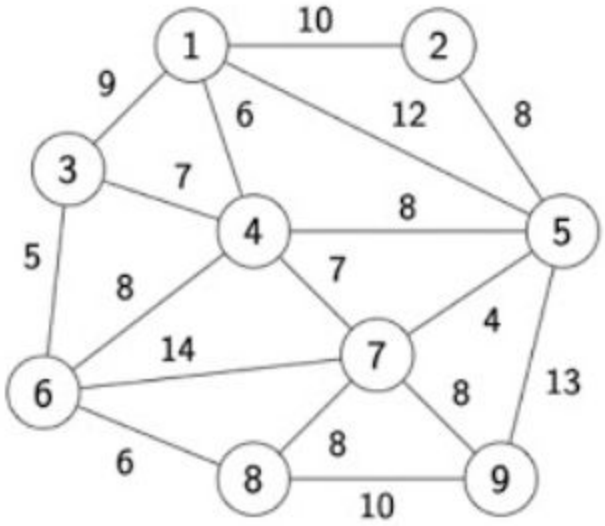
6


0
3
3
4
7
1
2
2

7

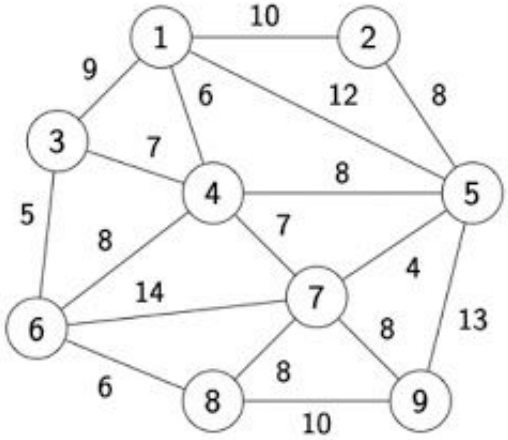

0
3
12
3
4
3
5
7
1
2
2

Minimum Spanning Tree

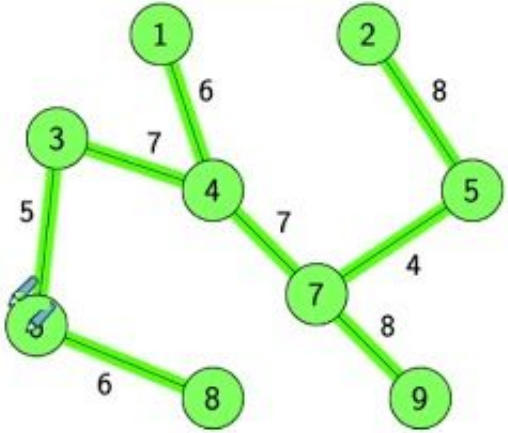


Given an undirected, edge-weighted graph, find a spanning tree (a tree involving all vertices) of the minimum total weight.

Minimum Spanning Tree

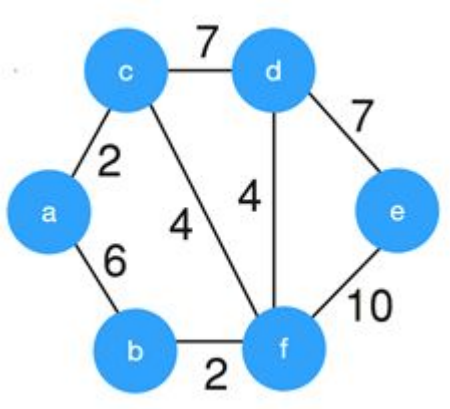


Total weight: 51

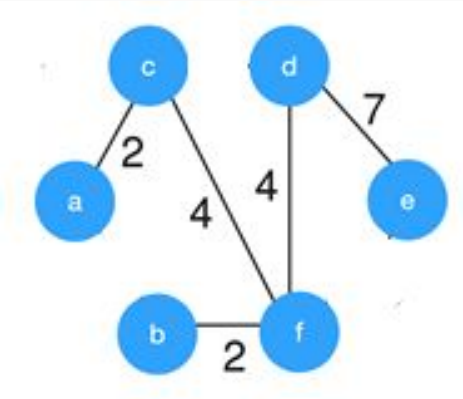


Given an undirected, edge-weighted graph, find a spanning tree (a tree involving all vertices) of the minimum total weight.

What is the weight of the minimum spanning tree using the Kruskal's algorithm?

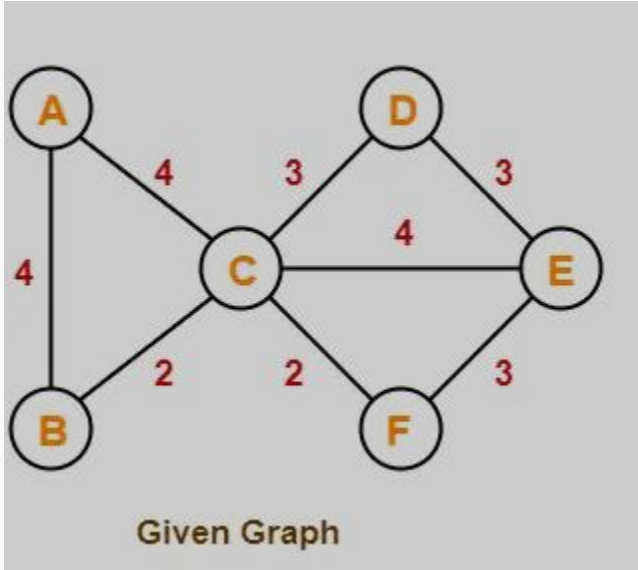


Solution on Kruskal's Minimum Spanning Tree (MST) Algorithm

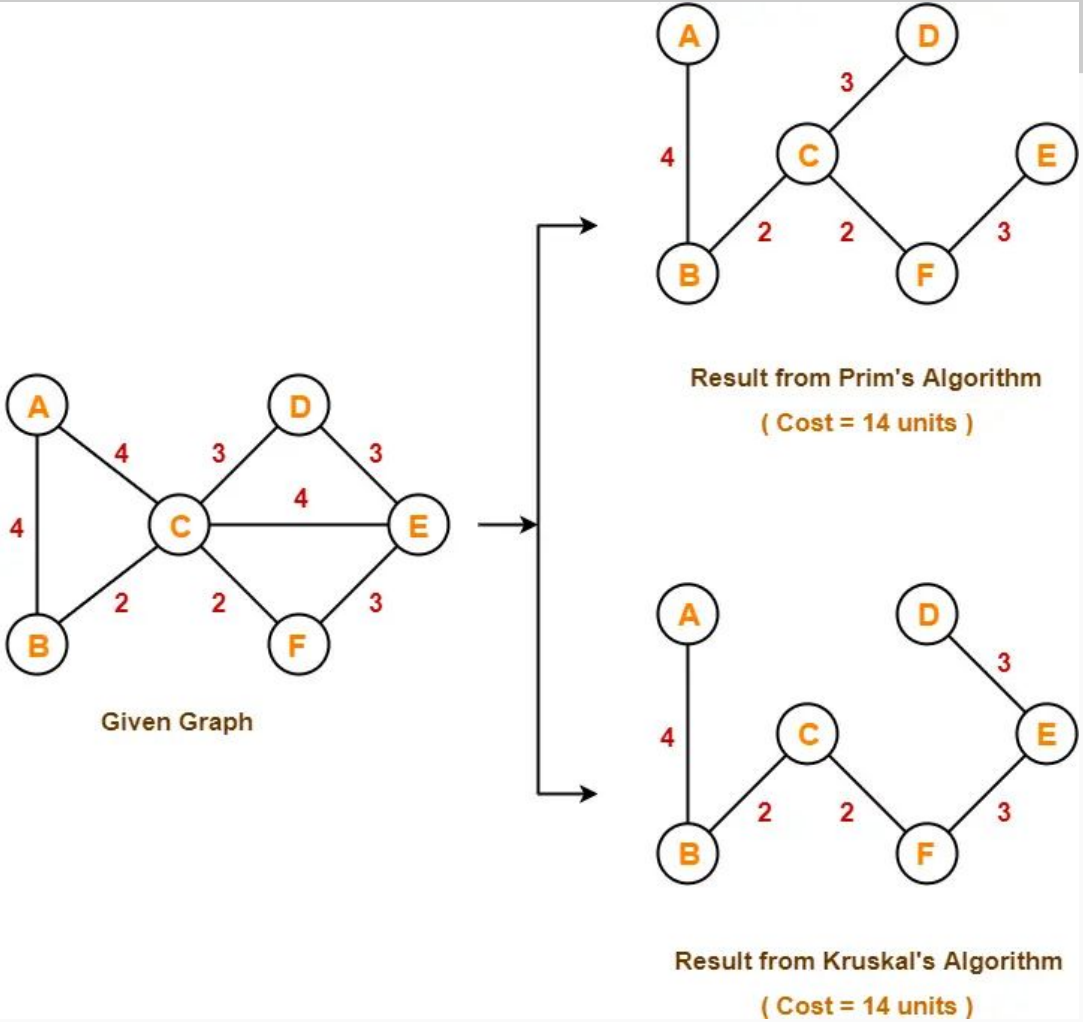


So, the weight of the MST is 19.

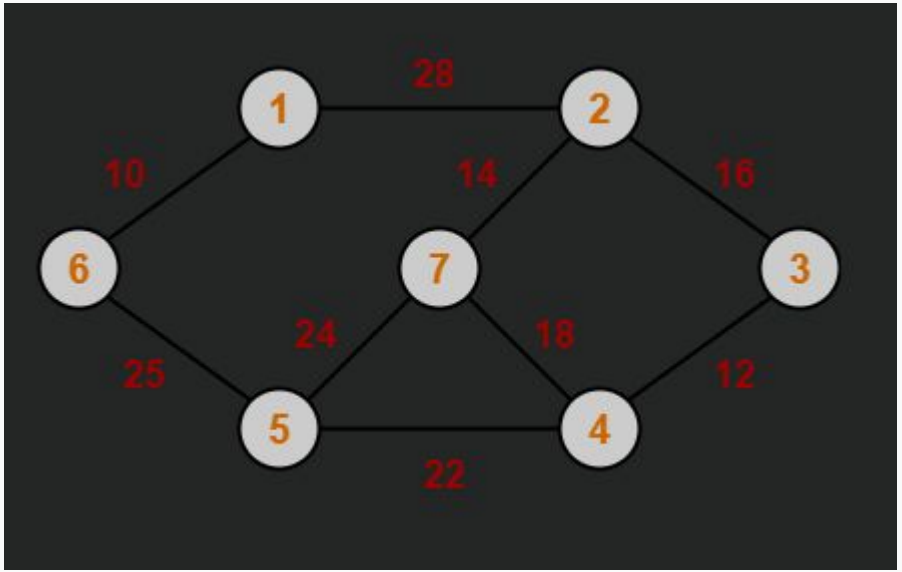
Questions on Kruskal's Minimum Spanning Tree (MST) Algorithm



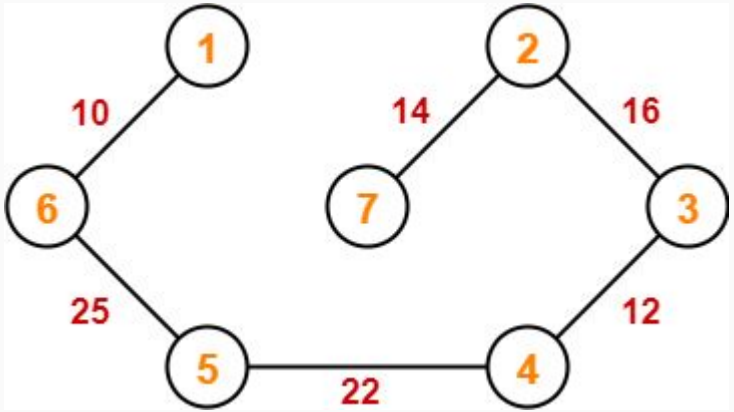
Solution


A
B
C
D
E
F
4
4
3
3
4
2
2
3
Given Graph
Result from Prim's Algorithm
(Cost = 14 units)
Result from Kruskal's Algorithm
(Cost = 14 units)

Questions on Kruskal's Minimum Spanning Tree (MST) Algorithm



Solution on Kruskal's Minimum Spanning Tree (MST) Algorithm



Since all the vertices have been connected / included in the MST, so we stop.

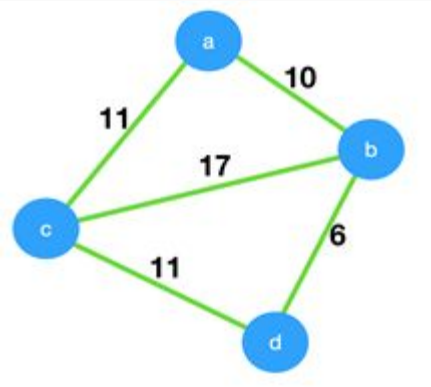
Weight of the MST

= Sum of all edge weights

= 10 + 25 + 22 + 12 + 16 + 14

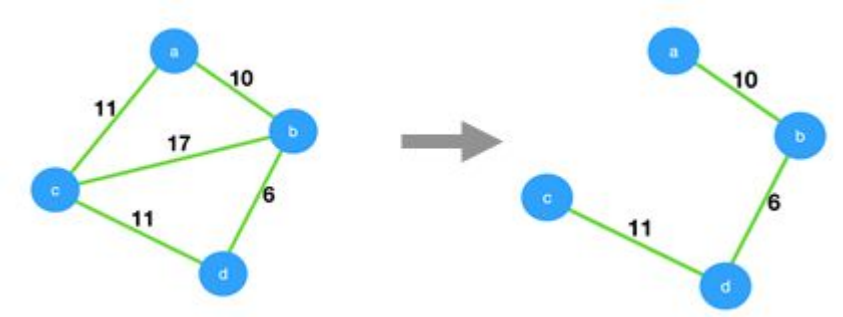
= 99 units

Questions on Prim’s Minimum Spanning Tree (MST) Algorithm



What is the weight of the minimum spanning tree using the Prim’s algorithm,starting from vertex a?

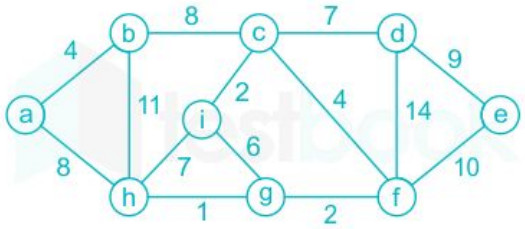
Solution



From, figure shown below weight of MST = 27

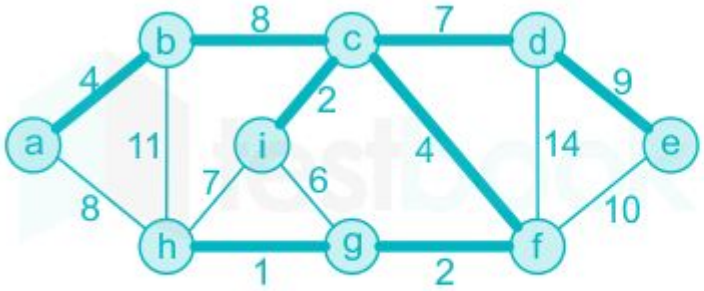
Questions on Prim's Minimum Spanning Tree (MST) Algorithm

Consider the undirected graph below:

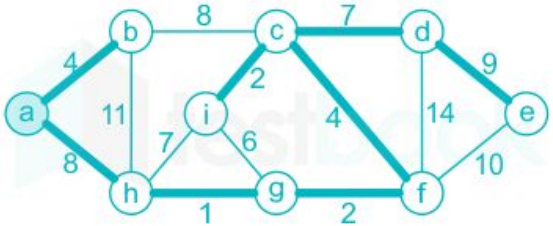


Using Prim's algorithm to construct a minimum spanning tree starting with node a, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

1. (a, b), (b, h), (g, h), (f, g), (c, f), (c, i), (c, d), (d, e)
2. (a, b), (b, h), (g, h), (g, i), (c, i), (c, f), (c, d), (d, e)
3. (a, b), (b, c), (c, i), (c, f), (f, g), (g, h), (c, d), (d, e)
4. (a, b), (g, h), (g, f), (c, f), (c, i), (f, e), (b, c), (d, e)
5. None of above



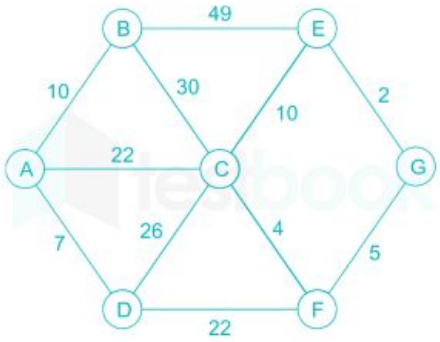
The final sequence will be (a, b), (b, c), (c, i), (c, f), (f, g), (g, h), (c, d), (d, e) with a cost 37.



Another final sequence is (a, b), (a, h), (g, h), (f, g), (c, f), (c, i), (c, d), (d, e) with a cost is 37(Two sequences possible).

Questions on Prim's Minimum Spanning Tree (MST) Algorithm

Consider the undirected graph below :

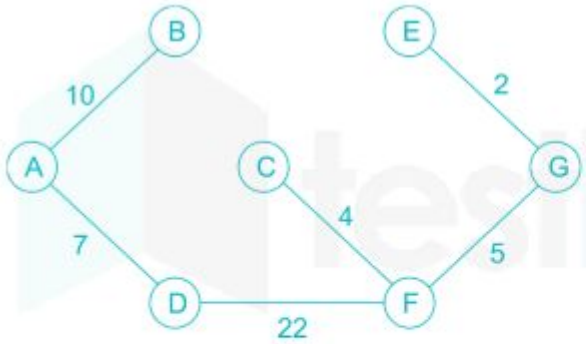


Using Prim's algorithm to construct a minimum spanning tree.

Which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree ?

1. (E,G) (C,F) (F,G) (A,D) (A,B) (A,C)
2. (E,G) (G,F) (F,C) (F,D) (D,A) (A,B)
3. (A,B) (A,D) (D,F) (F,G) (G,E) (F,C)
4. (A,D) (A,B) (D,F) (F,C) (F,G) (G,B)

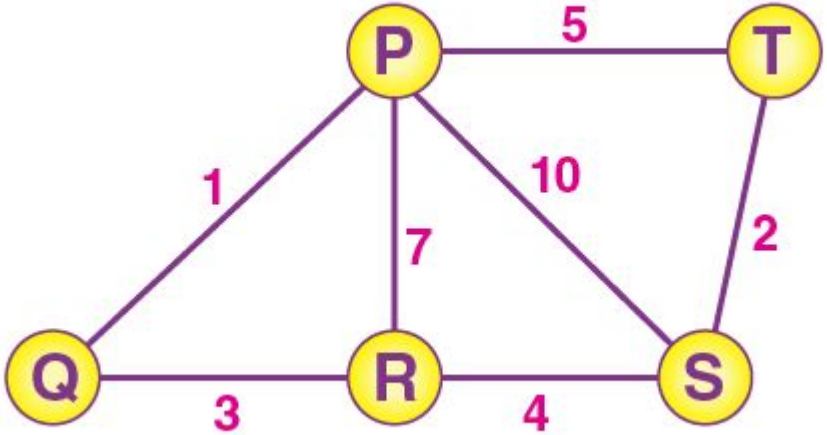
Solution

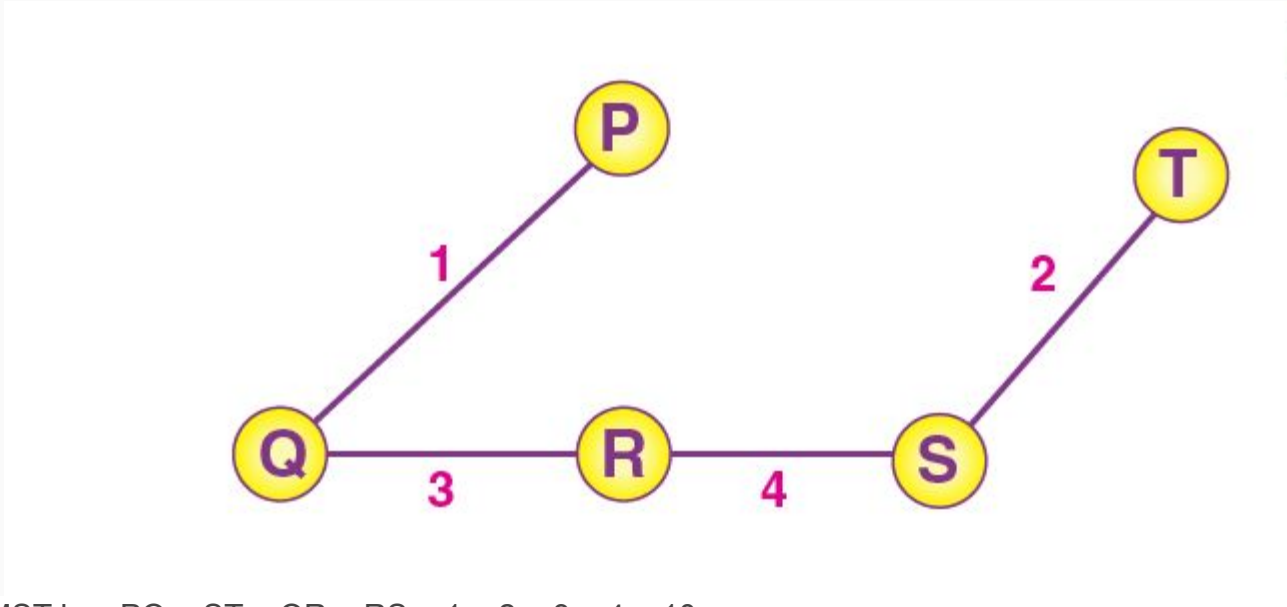


EG	E : ~~2~~, 10, 49
GF	G : ~~2~~, ~~5~~
FC	F : ~~4~~, ~~5~~, ~~22~~
FD	C : ~~4~~, 10, 22, 26, 30
DA	D : ~~7~~, ~~22~~, 26
AB	A : ~~7~~, ~~10~~, 22
	B : ~~10~~, 30, 49

So the order of adding edge is,

(E,G) (G,F) (F,C) (F,D) (D,A) (A,B)


P
T
Q
R
S
5
1
10
7
2
3
4



Total value of MST is = PQ + ST + QR + RS = 1 + 2 + 3 + 4 = 10.

Thank you!